CHAPTER 3

PRODUCTION AND COSTS

CHAPTER ESSENCE AT A GLANCE:
Understand short-run production relationships (Law of Variable Proportions), returns to scale, and the geometric behavior of cost curves (TC, TVC, TFC, AC, AVC, AFC, MC).

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Production Function and Time Horizons:** Production function specifies maximum output from factor inputs; Short Run contains fixed and variable inputs, whereas Long Run allows all inputs to vary.

• **Law of Variable Proportions:** As successive units of a variable factor are added to fixed factors, Total Product initially increases at increasing rate (Stage I), then increases at diminishing rate (Stage II), and eventually decreases (Stage III).

• **AC and MC Interrelationship:** Marginal Cost (MC) intersects Average Cost (AC) and Average Variable Cost (AVC) from below at their respective minimum points; when MC < AC, AC falls; when MC > AC, AC rises.

• **AFC Rectangular Hyperbola Nature:** Average Fixed Cost (AFC = TFC divided by Q) continuously declines as output expands because fixed overhead is spread over larger volume, forming a rectangular hyperbola that never touches either axis.

ESSENTIAL ECONOMIC MODELS & GOVERNING FORMULAS

Law of Variable Proportions $\Longleftrightarrow \mathrm{MP} = \frac{\Delta \mathrm{TP}}{\Delta L}$ and $\mathrm{AP} = \frac{\mathrm{TP}}{L}$

Marginal and Average Factor Productivity

Total Cost Composition $\Longleftrightarrow \mathrm{TC} = \mathrm{TFC} + \mathrm{TVC} \rightarrow \mathrm{AC} = \mathrm{AFC} + \mathrm{AVC}$

Short-Run Total and Unit Cost Partitioning

MC-AC Critical Geometry $\Longleftrightarrow \mathrm{MC} = \mathrm{AC}_{\min}$ and $\mathrm{MC} = \mathrm{AVC}_{\min}$

Intersection at Minimum Points of Unit Cost Curves

Marginal Cost Formula $\Longleftrightarrow \mathrm{MC}_n = \mathrm{TC}_n - \mathrm{TC}_{n-1} = \frac{\Delta \mathrm{TVC}}{\Delta Q}$

Incremental Cost of Expanding Output by One Unit

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Remember that Marginal Cost depends strictly on Total Variable Cost, not Total Fixed Cost; change in TC equals change in TVC because TFC is constant.

• **Alert:** AFC never touches the X-axis because TFC is always positive; AFC never touches the Y-axis because output cannot be zero when producing.

• **Alert:** Stage II of the Law of Variable Proportions is the only rational stage of production where a producer will operate.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Fixed Factors and Variable Factors of production with examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Fixed Factors:** Factor inputs whose quantity cannot be altered in the short run regardless of output level. Examples: Factory building, plant machinery, permanent staff. **[1 Mark]**

• **Variable Factors:** Factor inputs whose quantity can be readily varied in the short run directly with output changes. Examples: Raw materials, casual labour, electricity. **[1 Mark]**

2. Explain the Law of Variable Proportions with the help of a Total Product (TP) and Marginal Product (MP) schedule. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Statement:** When more units of a variable factor are combined with fixed factors, marginal product initially rises, then falls, and eventually becomes negative. **[1 Mark]**

• **Stage I (Increasing Returns):** TP increases at an increasing rate, MP rises and reaches maximum, ending where MP = AP. Caused by better division of labour and underutilized fixed capacity. **[1 Mark]**

• **Stage II (Diminishing Returns):** TP increases at a decreasing rate until it reaches maximum; MP is positive but falling until it becomes zero. Caused by imperfect factor substitutability. **[1 Mark]**

• **Stage III (Negative Returns):** TP begins declining; MP becomes negative. Caused by factor overcrowding and managerial diseconomies. Rational firm operates strictly in Stage II. **[1 Mark]**

3. Explain the relationship between Average Cost (AC) and Marginal Cost (MC) with a suitable diagram description. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **When MC < AC:** Average Cost falls continuously as output expands, pulling down the average. **[1 Mark]**

• **When MC = AC:** Average Cost is at its minimum; MC curve cuts the AC curve from below at its lowest point. **[1 Mark]**

• **When MC > AC:** Average Cost rises continuously as output expands; both curves are U-shaped, with MC reaching its minimum earlier than AC. **[1 Mark]**

4. The Total Fixed Cost of a firm is Rs 60. Given output (Q) and Total Cost (TC): Q=0, TC=60; Q=1, TC=80; Q=2, TC=96; Q=3, TC=120; Q=4, TC=160. Calculate TVC, AVC, and MC for each output level. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Formulae:** TVC = TC - TFC; AVC = TVC divided by Q; MC = TC(n) - TC(n-1). **[1 Mark]**

• **Calculations:**
• Q=1: TVC = 80 - 60 = Rs 20; AVC = 20 divided by 1 = Rs 20; MC = 80 - 60 = Rs 20.
• Q=2: TVC = 96 - 60 = Rs 36; AVC = 36 divided by 2 = Rs 18; MC = 96 - 80 = Rs 16. **[1.5 Marks]**

• • Q=3: TVC = 120 - 60 = Rs 60; AVC = 60 divided by 3 = Rs 20; MC = 120 - 96 = Rs 24.
• Q=4: TVC = 160 - 60 = Rs 100; AVC = 100 divided by 4 = Rs 25; MC = 160 - 120 = Rs 40. **[1.5 Marks]**

5. Why is the Average Fixed Cost (AFC) curve a rectangular hyperbola? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Mathematical Nature:** AFC equals Total Fixed Cost divided by Output (AFC = TFC divided by Q). Since TFC is a constant positive value, AFC declines continuously as output rises. **[1 Mark]**

• **Geometric Property:** The area under the AFC curve (AFC * Q = TFC) is identical at every point, making the curve an asymptotic rectangular hyperbola that approaches both axes without touching them. **[1 Mark]**

6. Distinguish between Returns to a Factor and Returns to Scale. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **Time Period:** Returns to a factor operate in the short run; Returns to scale operate strictly in the long run. **[1 Mark]**

• **Factor Proportions:** In returns to a factor, the proportion between fixed and variable factors changes; in returns to scale, all factor inputs are altered in the same constant proportion. **[1 Mark]**

• **Output Behavior:** Returns to a factor pass through increasing, diminishing, and negative stages; returns to scale exhibit constant, increasing, or decreasing returns to scale. **[1 Mark]**